Last time: For a a graph $G = (V, E)$, we defined:

(1) The **independence number** $\alpha(G)$ is the size of a largest independent set in G.

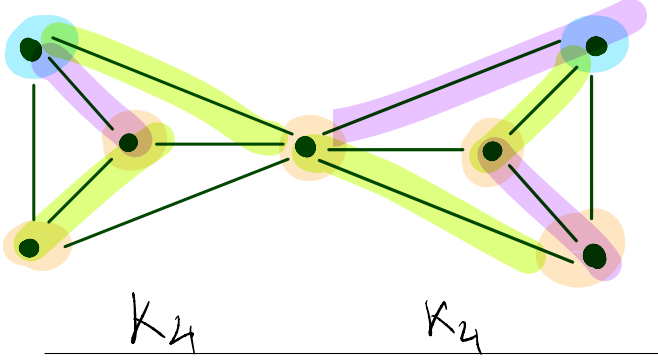
(2) The **edge-independence number** $\alpha'(G)$ is the size of a largest matching in G.

(3) A **vertex-cover** of G is a subset $U \subset V$ such that every edge of G has at least one end-point in U.
The **vertex-covering number** $\beta(G)$ is the size of a smallest vertex-cover of G.

(4) An **edge-cover** of G is a subset $S \subset E$ so that every vertex of G is incident to at least one edge in S.
The **edge-covering number** $\beta'(G)$ is the size of a smallest edge-cover of G.

→ only makes sense if $\delta(G) > 0$.

Example.



$\alpha(G) = 2$ (If we take three vertices two of them will lie on the same K_4)

$\alpha'(G) = 3$ ($\alpha' \leq \lfloor \frac{|V|}{2} \rfloor = 3$)

$\beta(G) = 5$

$\beta'(G) = 4$

Today:

- $\alpha'(G) \leqslant \beta(G) \leqslant 2\alpha'(G)$ for any graph.
- $\alpha'(G) = \beta(G)$ if G is bipartite.
- $\alpha'(G) + \beta'(G) = |V(G)|$ if G has no isolated vertices.
- $\alpha(G) + \beta(G) = |V(G)|$ for any graph (homework).

Theorem. *For any graph* G *we have* $\alpha'(G) \leq \beta(G) \leq 2\alpha'(G)$.

Proof: Let $M \subseteq E$ denote a maximum matching, so $\alpha'(G) = |M|$.

($\beta \leq 2\alpha'$) Let $C \subseteq V$ denote the set of endpoints of edges in M.

Then $|C| = 2|M|$ (by definition of matching)

Also, C must be a vertex cover, otherwise there is $e \in E$ with both endpoints not in C. Then $M \cup \{e\}$ is also a matching, contradicts maximality of M.

Hence $\beta(G) \leq |C| = 2|M| = 2\alpha'(G)$.

($\alpha' \leq \beta$) Let $C \subseteq V$ denote a minimum vertex cover, so $\beta(G) = |C|$.

Then each $e \in M$ has at least one endpoint in C, and these endpoints are distinct. Thus $|M| \leq |C|$, ie $\alpha'(G) = \beta(G)$.

Note: It is possible that $\beta(G) = 2\alpha'(G)$, just take $G = K_n$ with n odd.

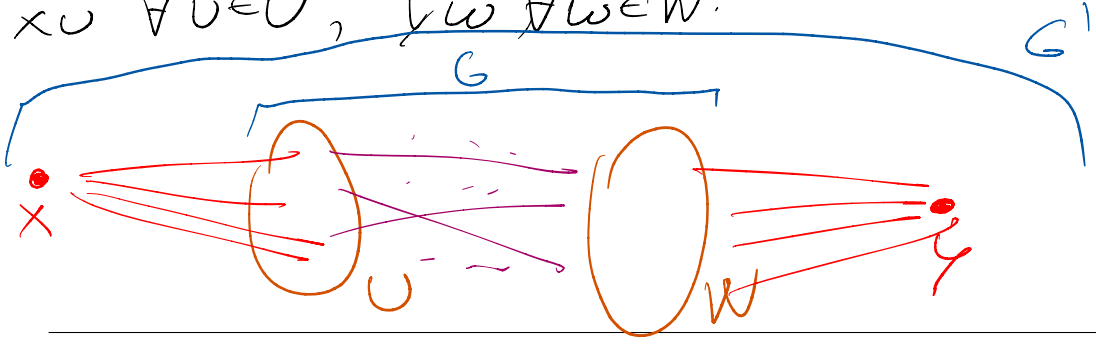
Next: $\alpha'(G) = \beta(G)$ is also possible. Bipartite!

Theorem (König's Theorem). *If G is a bipartite graph, then $\alpha'(G) = \beta(G)$. In other words, for bipartite graphs, the size of a maximum matching is equal to the size of a smallest vertex-cover.*

Recall: Menger's Thm:
Let G graph, x, y nonadjacent vertices. Then
|minimum x-y separating set| = maximum # of internally disjoint x-y paths.

Proof: (sketch) Let G bipartite with parts U, W.

Define G' by adding two vertices x, y to G, and edges
xu $\forall u \in U$, yw $\forall w \in W$.



Then

1) A matching in G corresponds to a set of internally disjoint x-y path in G'.

2) A vertex cover in G corresponds to an x-y cut in G'.

Thus $\beta(G) = |\text{minimum vertex cover for } G|$

$(2) = |\text{minimum } x\text{-}y \text{ CUT in } G'|$

Menger $=$ maximum # of internally disjoint x-y paths in G'

$(1) =$ maximum # of edges in a matching of G.

$= \alpha'(G)$.

∎

Lemma. *Let $G = (V, E)$ be a graph with no isolated vertices. If $S \subset E$ is a minimum edge-cover of G, then (V, S) is a forest with no isolated vertices.*

→ Graph with no cycles (i.e. disjoint union of Trees).

Proof: First: no isolated vertices in (V, S). Since S is an edge-cover, all vertices in V are incident to some edge in S.

second: Assume for a contradiction (V, S) contains a cycle.

say $(v_1, v_2, \dots, v_k, v_1)$ in (V, S), $k \geq 3$

Let $e = v_1v_2 \in S$. Then $S \setminus \{e\}$ is also an edge-cover of G:

Since v_1 is covered by $v_kv_1 \in S \setminus \{e\}$

v_2 is covered by $v_2v_3 \in S \setminus \{e\}$

Thus (V, S) contains no cycles ∎

$G = (V, E)$

Theorem (8.7). *If G is a graph on n vertices none of which is isolated, then* $\alpha'(G) + \beta'(G) = n$.

Idea: $\alpha' + \beta' \geq n$, $\alpha' + \beta' \leq n$

1) min. edge-covers S $\overset{?}{\rightsquigarrow}$ some matching M $\Rightarrow \alpha' \geq n - \beta'$
$|S| = \beta'$ with $|M| \geq n - |S|$ $\Rightarrow \alpha' + \beta' \geq n$

2) max. matching M $\overset{??}{\rightsquigarrow}$ some edge covers S $\Rightarrow \beta' \leq n - \alpha'$
$|M| = \alpha'$ with $|S| \leq n - |M|$ $\alpha' + \beta' \leq n$

Proof ($\alpha' + \beta' \leq n$). Let M maximum matching, so $|M| = \alpha'$

Let $A \subseteq V$ set of endpoints of M, let $B = V \setminus A$

Now, for each $b \in B$ (non isolated) choose some edge $e \in E$ incident to b, Thus $e \notin M$ (def^n of B). These edges form a set $S \subseteq E \setminus M$ with $|S| \leq |B|$ which covers B

Since $V = A \sqcup B$, we see that $M \cup S$ is an edge-cover

(A covered by M; B covered by S)

$|B| = n - |A| = n - 2\alpha'$

Thus $\beta'(G) \leq |M \cup S| = |M| + |S| = \alpha'(G) + |S| \leq \alpha'(G) + |B|$

$= \alpha'(G) + n - 2\alpha'(G)$

$= n - \alpha'(G)$

Hence $\alpha'(G) + \beta'(G) \leq n$

($\alpha' + \beta' \geq n$) Let $S \subseteq E$ minimum edge-cover, so $|S| = \beta'(G)$

By Lemma, (V, S) is a forest, with no isolated vertices

Let $G_1, \dots, G_k$ denote the components of (V, S)

Recall: (V, S) has $n-k$ edges ($|S| = |V| - \#\text{components}$)

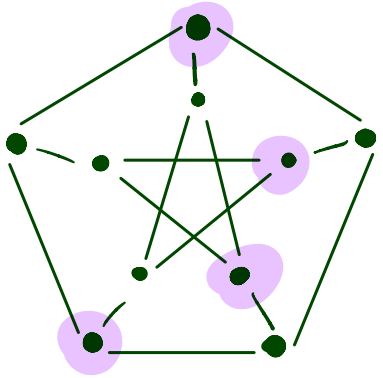
Thus $|S| = n - k$. Since no vertex in (V, S) is isolated, we select one edge in G_i for each G_i, and this gives a matching M in G with $|M| = k$. Thus $\alpha'(G) + \beta'(G) \geq |M| + \beta'(G)$

$= |M| + |S| = k + (n-k) = n$

i.e. $\alpha'(G) + \beta'(G) \geq n$ ☒

Theorem (8.8). *If* G *is a graph on* n *vertices, then* $\alpha(G) + \beta(G) = n$.

Example. *Compute* $\alpha(P)$ *and* $\beta(P)$ *where* P *is the Petersen graph.*



$\alpha(P) = ?$

We have $\alpha' \geq 4$

And if we take 5 vertices, Then 3 of Those will lie either on The inner C_5, or The outer C_5

Thus $\alpha(P) = 4$

Hence $\beta(P) = 10 - 4 = 6$

Next: a nice consequence of Hall's theorem.

Theorem. *Let* G *be a bipartite graph with parts* U, W *such that* $|U| \leqslant |W|$. *Assume that* G *has no isolated vertices and that* $\deg u \geqslant \deg w$ *whenever* $u \in U$ *and* $w \in W$ *are such that* $uw \in E(G)$. *Then* G *has a matching which saturates* U.

(Thus $\alpha' = |U|$)

Proof: Can verify Hall's condition: $|N(S)| \geq |S| \quad \forall \emptyset \neq S \subseteq U$

